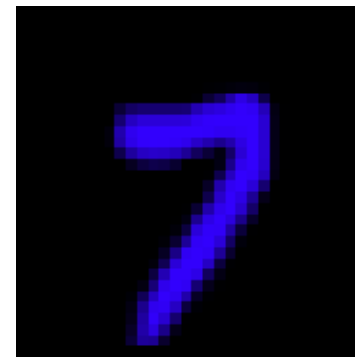
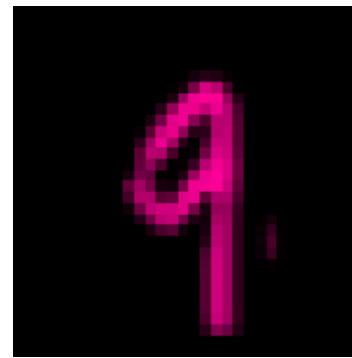
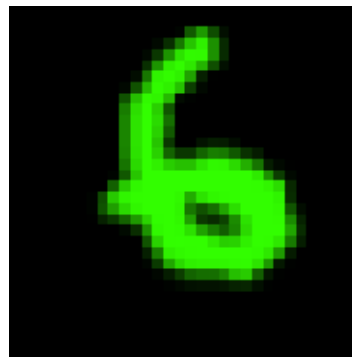
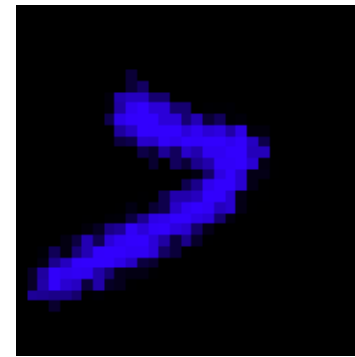
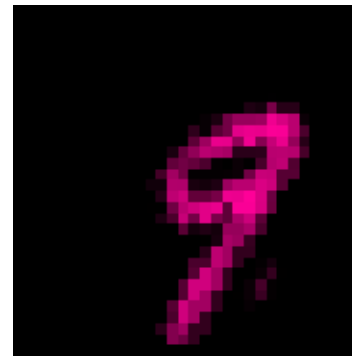
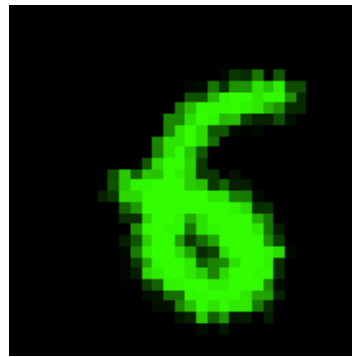
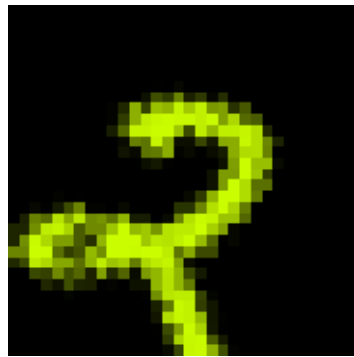
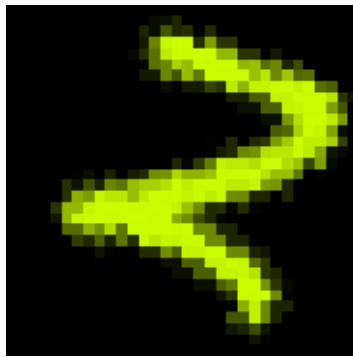


CMNIST Shift Visuals (Rot/Light/Huber)

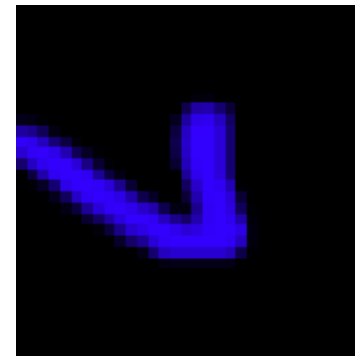
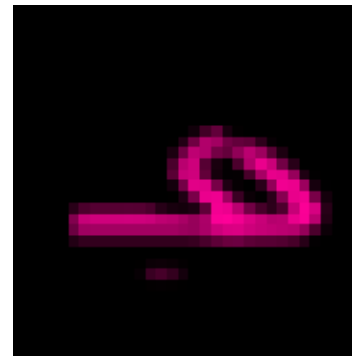
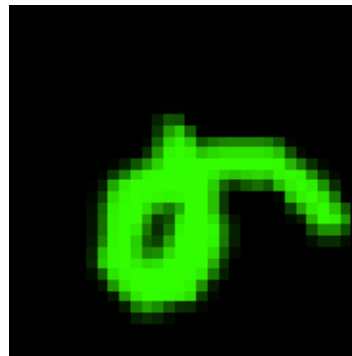
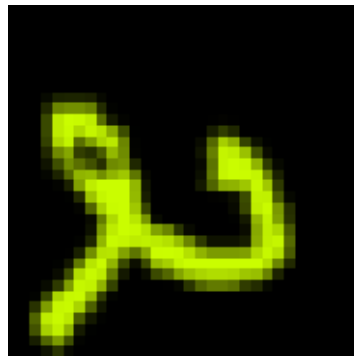
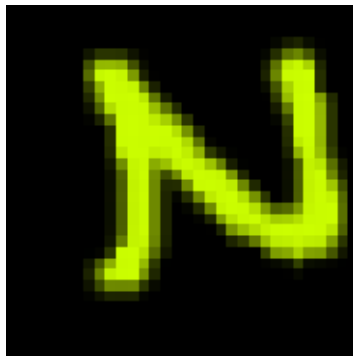
Original
(Huber 0)



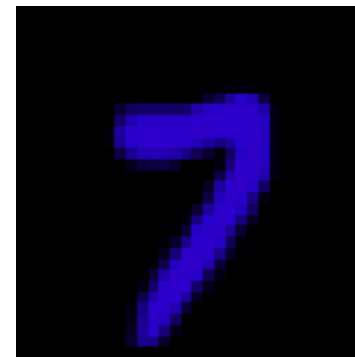
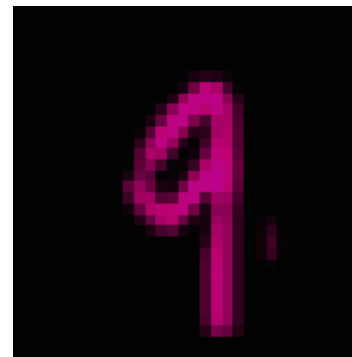
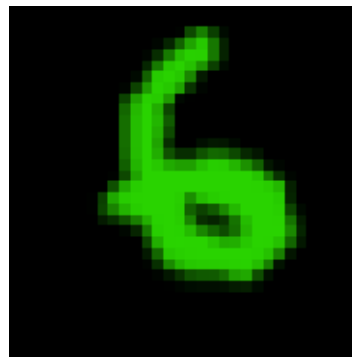
Rotation (LOW)
30.0 deg



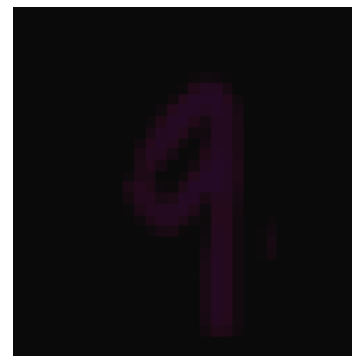
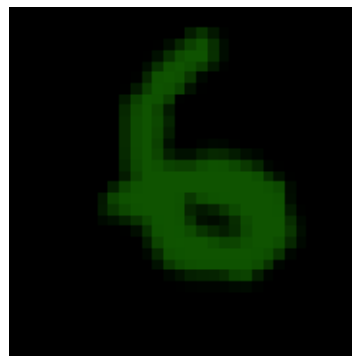
Rotation (HIGH)
90.0 deg



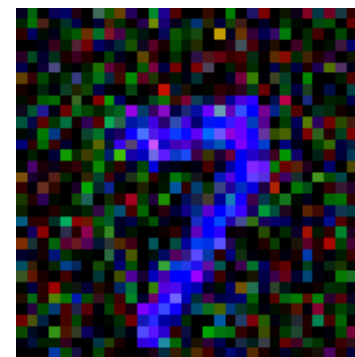
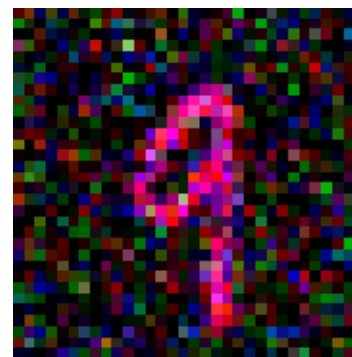
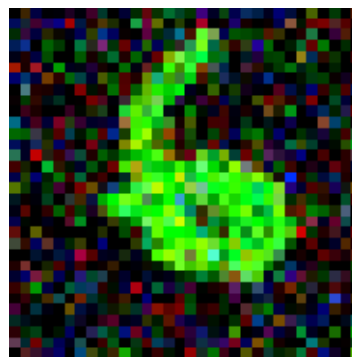
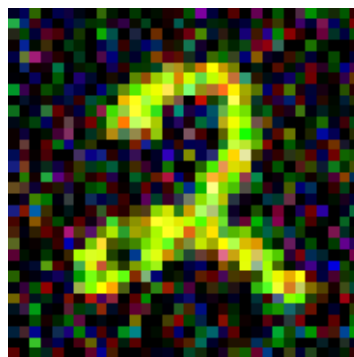
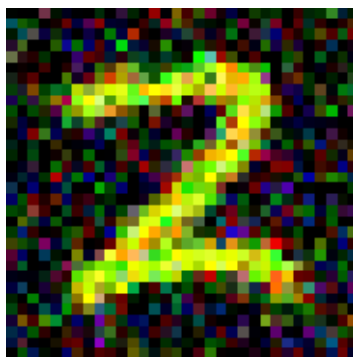
Lighting (LOW)
Jitter=0.3



Lighting (HIGH)
Jitter=1.0



Huber
(Alpha=1.0)



Relative Error: Rotation

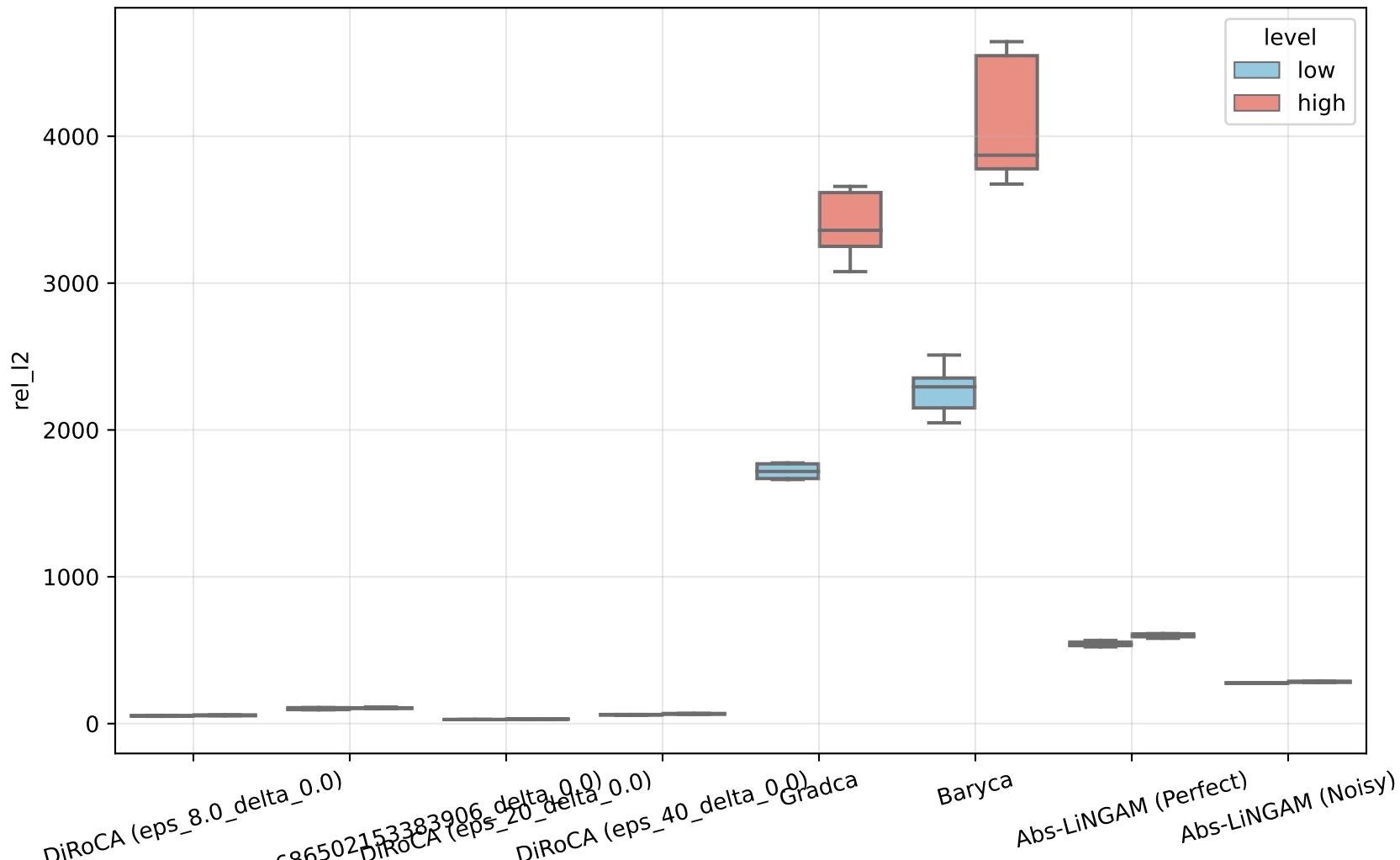


Table: Rotation

Method	Low	High
Abs-LiNGAM (Noisy)	276.60	288.02
Abs-LiNGAM (Perfect)	543.27	608.37
Baryca	2287.77	4096.16
DiRoCA (eps_20_delta_0.0)	27.99	30.74
DiRoCA (eps_40_delta_0.0)	61.44	66.47
DiRoCA (eps_61.686502153383906_delta_0.0)	101.67	106.30
DiRoCA (eps_8_0_delta_0.0)	52.88	56.68
Gradca	1717.83	3624.06

Relative Error: Lighting

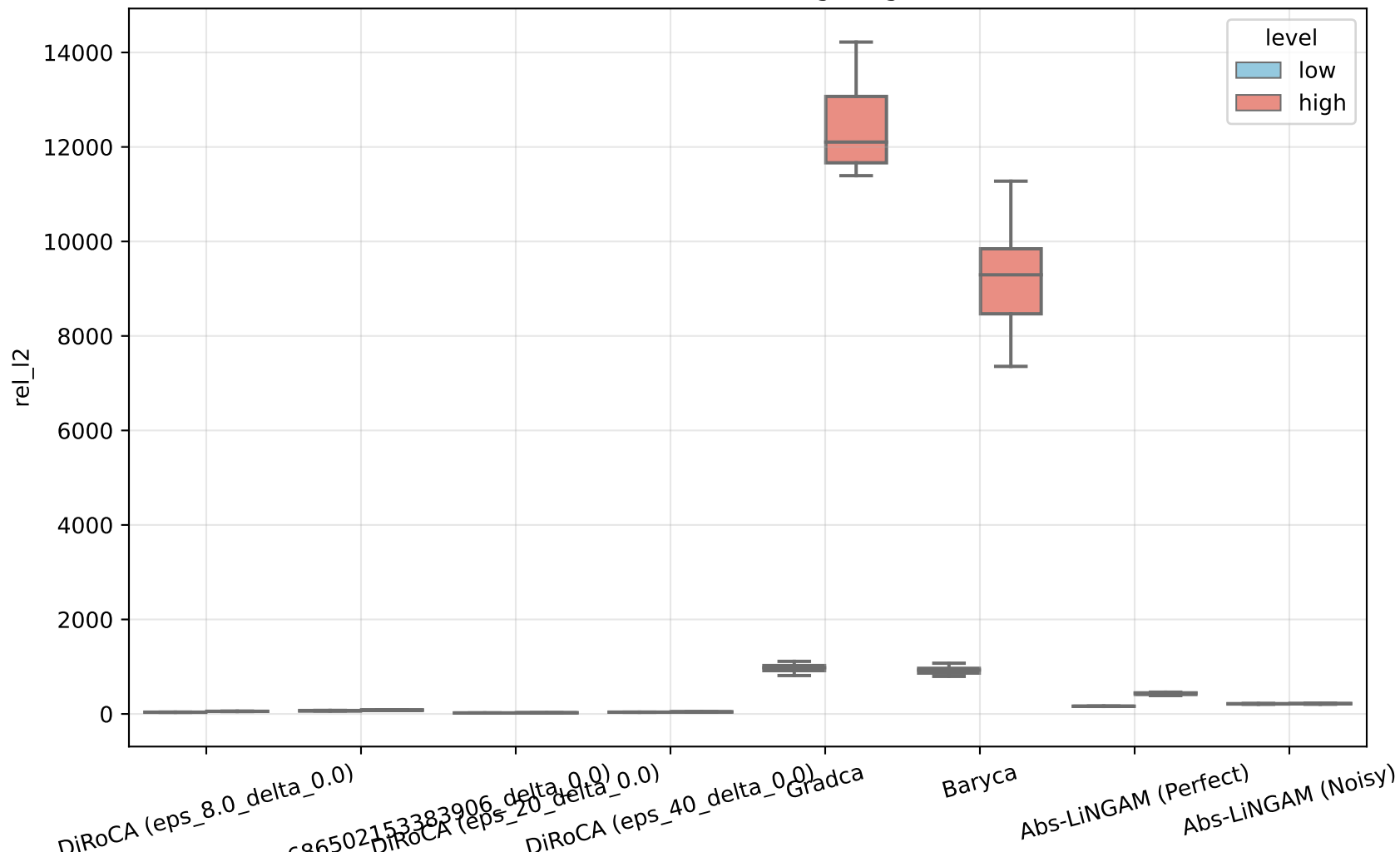


Table: Lighting

Method	Low	High
Abs-LiNGAM (Noisy)	213.84	217.37
Abs-LiNGAM (Perfect)	163.33	436.40
Baryca	920.45	9221.21
DiRoCA (eps_20_delta_0.0)	20.41	24.32
DiRoCA (eps_40_delta_0.0)	37.98	43.04
DiRoCA (eps_61.686502153383906_delta_0.0)	67.66	80.38
DiRoCA (eps_8_0_delta_0.0)	36.33	55.28
Gradca	967.38	12424.55

Huber Contamination

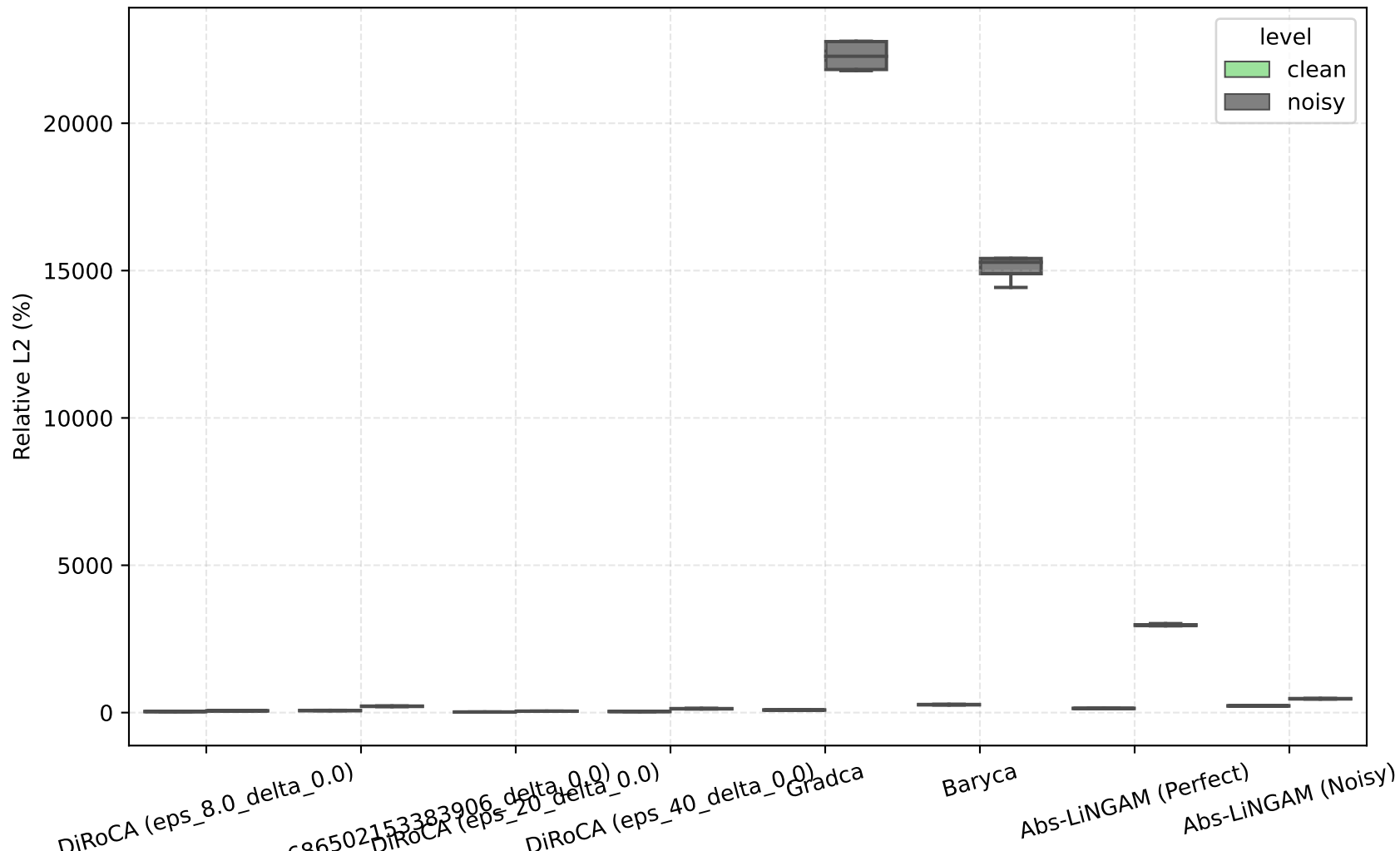


Table: Huber Contamination

Method	Clean ($\alpha=0$)	Noisy ($\alpha=1$)
Abs-LINGAM (Noisy)	227.89	470.19
Abs-LINGAM (Perfect)	141.64	2956.93
Baryca	270.19	15326.38
DiRoCA (eps_20_delta_0.0)	20.05	48.15
DiRoCA (eps_40_delta_0.0)	38.68	132.71
DiRoCA (eps_61.686502153383906_delta_0.0)	69.80	215.12
DiRoCA (eps_8.0_delta_0.0)	35.97	60.37
Gradca	87.64	22265.58